



Exercise 1 - How much should you trust an LLM about a dataset?

Large language models such as ChatGPT or Claude can be very useful coding and learning assistants. They are also capable of giving precise-sounding answers that are partly or completely wrong. In this exercise, you will test an LLM on a factual question and compare its claims with the dataset itself.

1. Ask an LLM to describe the bicycle rental dataset used in this course. Ask specifically for the column names, the possible values or ranges of the variables, the number of observations, and the response variable. Keep a copy of the answer it gives you.
2. Load the OpenML version of the bicycle rental dataset (data_id = 42712) using the code below.

```
from sklearn.datasets import fetch_openml

bikes = fetch_openml(
    data_id=42712, as_frame=True, parser="auto"
).frame
```

3. Choose three concrete claims from the LLM response and verify them directly from the data. Prefer claims that can be checked exactly - for example, a row count, a column name, a minimum or maximum value, or whether missing values are present. For each claim, report whether it was correct and show the line of Python code that allowed you to decide.
4. Reflect on the result. If the model made an error, would you have noticed it without checking? Which kinds of statements did it handle well, and which kinds of statements should you verify directly from the data before using them in an analysis or report?

Exercise 2 - Two baselines for five-hour wind forecasting

Before fitting a machine-learning model, establish what very simple predictions can already achieve. Here you will compare two no-learning baselines for predicting the Luzern wind peak five hours into the future.

1. Load the weather dataset and construct the five-hour-ahead response vector as follows:

```
import pandas as pd

weather = pd.read_csv(PATH_TO_YOUR_DATA)

y = weather["LUZ_wind_peak"][5:].values
```

2. **Mean baseline.** Predict the same value for every observation: the mean of y . Compute and report the mean squared error (MSE).
3. **Persistence baseline.** Predict that the wind peak five hours from now will be equal to the wind peak observed now: `weather["LUZ_wind_peak"][:-5].values`. Compute and report the MSE.
4. Compare the two MSE values. Which baseline is better, and by how much? Keep both numbers: any more sophisticated model should improve on these baselines to be useful.

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5. The persistence forecast may seem like the obvious choice, yet at a five-hour horizon it performs worse. Explain why. At what forecast horizon would you expect persistence to become the stronger baseline?
 6. Compare the MSE of the mean predictor with `y.var()`. They are identical. Explain why this equality is expected rather than a coincidence.

Exercise 3 - Classification baselines: when accuracy can mislead

The same baseline idea applies to classification, but here the response is a class label rather than a number. In this exercise, you will use the spam dataset to see why accuracy must always be interpreted in relation to the class balance and the kinds of mistakes a classifier makes.

1. Load the spam dataset and calculate the proportion of emails labelled as spam.

```
import pandas as pd

spam = pd.read_csv(
    "PATH_TO_YOUR_DATA", nrows=6000
).dropna()
```

2. Consider a trivial classifier that predicts “ham” for every email, regardless of its content. What accuracy does this classifier achieve? Treat this as a baseline that a useful spam filter should improve upon.
3. Suppose a spam filter achieves 90% accuracy on this dataset. Is that a meaningful improvement over the baseline? Now imagine a different mailbox in which only 3 out of every 100 emails are spam. Would 90% accuracy still be impressive? Explain your reasoning using the class proportions.
4. Two spam filters have exactly the same overall accuracy. Filter A occasionally lets spam reach the inbox; Filter B occasionally deletes legitimate email. Are these errors equally important? State what additional performance measures you would want reported instead of accuracy alone, and explain why they are more informative.